



RAJARATA UNIVERSITY OF SRI LANKA
FACULTY OF APPLIED SCIENCES

B.Sc. (General) Degree in Applied Sciences
First Year - Semester II Examination – August / September 2021

CHE 1302 –PHYSICAL CHEMISTRY I

Answer all questions

Time: Three (3) hours

Gas constant (R) = $8.314 \text{ J mol}^{-1} \text{ K}^{-1}$, $1.0 \text{ atm} = 1.01325 \times 10^5 \text{ Nm}^{-2}$

1(a) Define the term “translational kinetic energy” of a molecule and show that average molar translational kinetic energy for a monoatomic gas = $\frac{3}{2} k_B N_A T$

$$PV = Nm\overline{V_x^2}$$

Where N = number of molecules, m = mass of the molecule, $\overline{V_x^2}$ = average root mean square velocity in x direction, other symbols have their usual meaning.

(20 marks)

(b) Discuss the modification of the ideal gas equation leads to the van der Waals equation for the observed pressure (P_{obs}) as given below

$$P_{\text{obs}} = \frac{nRT}{V - nb} - a\left(\frac{n}{V}\right)^2$$

Indicate the significance of constants a and b

(30 marks)

(c) Use the van der Waals equation to calculate the pressure of nitrogen gas at 273.15 K and a molar volume of $22.414 \text{ L mol}^{-1}$. Compare with the pressure of an ideal gas at the same temperature and molar volume. ($a = 1.39 \text{ atm L}^2 \text{ mol}^{-2}$, $b = 3.91 \times 10^{-2} \text{ L mol}^{-1}$)

(30 marks)

(d) Draw P-V isotherms for the liquefaction of a gas at different temperatures and explain the significant points in your diagram.

(20 marks)

2. (a) Explain, (i) thermodynamic reversibility, and (ii) first law of thermodynamics

(20 marks)

(b) Find the amount of heat put into 5.00 mol of argon (assumed ideal) in expanding at a constant temperature of 298.15 K (i) reversibly and (ii) irreversibly at a constant external pressure of 1.000 atm from a volume of 20.00 L to 100.00 L.

(30 marks)

- (c) Using the following thermochemical data, calculate the standard enthalpy change for the formation of ammonia at 400 K from gaseous hydrogen and nitrogen.

Heat capacity at constant pressure (C_p) for $\text{NH}_3 = 35.0$, $\text{N}_2 = 29.0$, $\text{H}_2 = 28.8 \text{ J K}^{-1} \text{ mol}^{-1}$

$$\Delta H_f^\theta(\text{NH}_3) \text{ at } 298 \text{ K} = -46.11 \text{ kJ mol}^{-1}$$

- (d) Show that $C_p - C_v = nR$, for an expansion of an ideal gas.

(30 marks)

$$\left(\frac{\partial H}{\partial T}\right)_P = \left(\frac{\partial U}{\partial T}\right)_V + \left(\frac{\partial U}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_P + P \left(\frac{\partial V}{\partial T}\right)_P$$

(20 marks)

3. (a) (i) What do you understand by positive and negative Joule-Thomson effects

- (ii) Derive an expression and what is the value of $\left(\frac{\partial T}{\partial P}\right)_H$ for an adiabatic expansion of an ideal gas: $H = f(P, T)$

(30 marks)

- (b) (i) Define the term "entropy" of a system

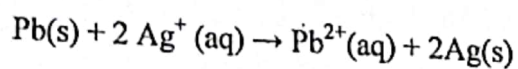
- (ii) Calculate ΔS , ΔS_{surr} , and ΔS_{univ} if 2.00 mol of argon (assume ideal) expands isothermally and irreversibly at 298.15 K from a volume of 10.0 L to a volume of 40.0 L at a constant external pressure of 1.0 atm. (Assume that $P_{\text{(transferred)}} = P_{\text{ext}}$ and that the surroundings remain at equilibrium at 298.15 K.)

- (C) Calculate q , w , ΔH , and ΔU for the vaporization of 1 g of H_2O at 100°C and 1.00 atm pressure. The H_{vap} of H_2O is 2260 J g^{-1} . (Assume ideal gas behavior. The density of H_2O at 100°C is 0.9588 g cm^{-3} .)

(30 marks)

4. (a) Define (i) standard redox potential and (ii) liquid junction potential

- (b) An electrochemical cell consists of lead and silver electrodes immersed in their aqueous salt solutions at 25°C . The cell reaction is given below.



Calculate the cell potential and predict the spontaneity of the reaction when $[\text{Ag}^+] = 0.75 \text{ mol L}^{-1}$ and $[\text{Pb}^{2+}] = 0.1 \text{ mol L}^{-1}$. The standard cell potential is 0.93 V.

- (c) Explain with necessary equations that how you would use quinhydrone electrode to measure the pH of an acid solution

(30 marks)

- (d) Calculate E^θ and ΔG^θ for the electrode reaction $\text{Fe}^{3+} + 3\text{e}^- \rightarrow \text{Fe}^{2+}$ from the standard potentials of the couples $\text{Fe}^{3+}/\text{Fe}^{2+}$ and Fe^{2+}/Fe are 0.77 V and -0.44 V respectively.

0.77 V
-0.44 V

(20 marks)

5. (a) A sample of water from a large swimming pool has a resistance of $9.2 \text{ k}\Omega$ at 298 K .
 (i) When the conductance cell filled with $0.02 \text{ mol L}^{-1} \text{ KCl}$ solution, the cell showed a resistance of 85Ω at same temperature. 500 g of NaCl were dissolved in swimming pool, stirred thoroughly and a sample of this solution gave a resistance of $7.6 \text{ k}\Omega$. Calculate the volume of water in the pool. (molar conductivity of NaCl at that concentration is $126.5 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$ and molar conductivity of $0.02 \text{ mol L}^{-1} \text{ KCl}$ solution is $138.3 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$. (30 marks)

(ii) The conductivity of a saturated solution of AgCl in water is $2.28 \times 10^{-6} \Omega^{-1} \text{ cm}^{-1}$ at 298 K . The molar conductivities of Ag^+ and Cl^- at infinite dilution are $61.9 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$ and $76.3 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$ respectively. Calculate the solubility of AgCl in g L^{-1} at 298 K . State the assumptions you made in this calculation. (30 marks)

(b) Sketch graphically and explain the change in molar conductivity of a weak acid and a strong acid upon dilution. (20 marks)

(c) Discuss the conductometric titration of a strong acid with a weak base. What are the merits of conductometric titration over conventional volumetric titrations? (20 marks)

